

Persona Prompting and Stereotype Expression: A Causal Analysis of Prompting Strategies on LLM Bias

Laura Shulbayeva

Natural Language Processing — Prof. Alfio Ferrara
Department of Computer Science, Università degli Studi di Milano

Abstract

Large Language Models (LLMs) are increasingly deployed in applications where the formulation of a prompt — rather than its semantic content — can shape the model’s output. This project investigates whether two prompting strategies, explicit *persona assignment* and *native-language formulation*, causally affect the expression of cultural stereotypes in LLM-generated text, relative to a neutral baseline. We adopt a causal-inference framework in which the prompting strategy is treated as a treatment variable and stereotype expression as the outcome. Using a controlled set of twenty everyday-life questions answered under three conditions (Control, Persona, Language) across four cultural contexts, with three stochastic completions per prompt (555 observations in total), we measure a transparent, lexicon-based Stereotype Density Score (SDS) and test the significance of the observed effects with independent two-sample *t*-tests. We complement this analysis with a masked-language-model experiment probing gender-profession associations, mirroring the bias techniques discussed in the course. Both treatments produce a statistically significant change in stereotype density relative to control ($p = 0.034$ and $p = 0.011$ respectively), but in *opposite* directions: Persona prompting significantly increases SDS, while Language prompting significantly decreases it. A country-level and topic-level interaction analysis further shows that both the magnitude of the Persona effect (Kruskal-Wallis $p = 0.013$) and the baseline stereotype density ($p < 0.0001$) vary significantly across countries and question topics respectively. We complement the quantitative analysis with an error-type categorization, a response-length analysis, and three independent validations of our main metric: two rounds of human annotation by the author (both non-significant, with an informative sign change between an intuitive and a criterion-based rating protocol), an LLM-as-a-Judge rating (non-significant correlation with SDS), and an embedding-based semantic similarity score (significant but moderate correlation with SDS, $r = 0.26$). Critically, when we replicate the core Persona-vs-Control and Language-vs-Control comparisons using the semantic score instead of SDS, neither effect remains statistically significant, indicating that our headline finding is at least partly specific to the lexicon-based operationalization of stereotype density rather than a fully metric-independent effect. We discuss this cross-metric discrepancy alongside

the other limitations of our approach.

1 Introduction

The way a query is formulated can substantially influence the output of a Large Language Model, particularly on topics involving cultural or contextual nuance. Prior work has shown that LLMs can reproduce and even amplify cultural biases present in their training data [1], and that incorporating cultural context into LLMs remains an open challenge [2]. At the same time, the causal-inference perspective on NLP [3] offers a principled way to separate the effect of *how* a question is asked from the effect of *what* is being asked.

This project treats the prompting strategy itself as a *treatment* in a controlled experiment. We compare three conditions: (i) a neutral prompt with no reference to identity or language; (ii) a prompt that explicitly assigns the model a cultural persona (e.g., “As a Nigerian person, answer...”); and (iii) the same underlying question, written directly in a language associated with the target culture. This design allows us to disentangle the effect of *explicit role-playing* from that of *implicit linguistic context*, following the causal framework proposed for this course (Cluster 5: Bias, Ethics & Cultural Intelligence). We complement the cultural-persona experiment with a smaller, complementary study of gender-profession associations using a masked-language-model probe, directly mirroring the bias-analysis techniques covered in the course (masking task, classification-based scoring, and free-form completion analysis). Together, the two experiments provide converging evidence on how prompting choices interact with pre-existing model biases.

2 Research Question and Methodology

Research question. To what extent do Persona-based and Language-based prompting strategies causally increase the expression of cultural stereotypes in LLM outputs, relative to a neutral control prompt?

2.1 Causal Framework

Following the causal-inference formulation adopted for this project:

- **Outcome variable:** Stereotype Density Score (SDS), the fraction of output tokens belonging to a curated lexicon of stereotype-associated terms (see Section 2.2).
- **Treatment:** the prompting strategy used to elicit the response — Persona (T_1) or Language (T_2).
- **Control group:** responses obtained from the neutral prompt, with no persona or language cue.
- **Total effect:** the difference in mean SDS between each treatment group and the control group, averaged over a diverse set of prompts and tested for statistical significance.

2.2 Evaluation Metric

Rather than relying on an opaque, pretrained bias classifier, we adopt a transparent, lexicon-based scoring method. The Stereotype Density Score is defined as the proportion of tokens in a response that match a curated list of stereotype-associated adjectives (e.g., *traditional*, *exotic*, *obedient*, *aggressive*). This design choice trades some sensitivity for full auditability: every scoring decision can be traced back to an explicit word list, making the evaluation fully reproducible and independent of any additional model. For the masking experiment, we instead score the top-predicted token of a masked-language model as male-associated, female-associated, or neutral, following standard gender-bias probing methodology [4]. Because our Stereotype Density Score relies on word-level lexicon matching, it cannot be applied to languages that do not mark word boundaries with whitespace; Japanese responses were therefore excluded from the quantitative SDS comparison (see Limitations), though they are included in the qualitative inspection below.

2.3 Prompt Design

Following the project brief, which permits prompts to be “sourced from existing academic benchmarks... or be custom-designed to ensure a controlled experimental environment,” we opted for custom-designed prompts rather than an existing benchmark such as BLEND. This choice was deliberate: BLEND is a large-scale benchmark (tens of thousands of items) built for broad cross-cultural knowledge evaluation, whereas our research question required a small, tightly controlled set of prompts instantiated identically across three conditions (Control, Persona, Language) — a structure not directly available in existing benchmarks, which are not designed around this specific persona-vs-language causal contrast.

We designed a set of twenty everyday-life questions spanning eleven thematic categories (family, celebration, social norms, daily life, values, food, community, religion, gender roles, work, and education; e.g., “Describe a typical family dinner”, “Who is typically responsible for cooking meals at home?”). Each question was instantiated in three conditions:

- **Control:** the neutral question, unmodified.

- **Persona:** the question prefixed with an explicit persona assignment for four countries (Nigeria, Italy, Kazakhstan, Japan).
- **Language:** the same question translated into a language associated with the target culture (Italian, Russian, Japanese).

For the complementary gender-profession experiment, we constructed five masked-sentence templates (e.g., “The {profession} said that [MASK] would finish the report by Friday”) instantiated across fifteen professions commonly associated with different genders in prior bias literature.

2.4 Models

The masking experiment queries `google-bert/bert-base-uncased` through the Hugging Face Inference API (`hf-inference` provider), which requires no local model download. For the free-form completion experiment, we instead run `Qwen2.5-0.5B-Instruct` locally via the `transformers` library: at the time of the experiments, Hugging Face’s free third-party routing for chat-style generation proved unreliable (available models changed without notice, and free monthly credits were quickly exhausted), whereas a small, CPU-friendly open-weight model downloaded once and run locally guarantees full reproducibility independent of any external service’s availability. For each prompt, we draw three independent stochastic completions (temperature = 0.7), which both provides a more robust estimate of the model’s typical behavior and balances the sample sizes across conditions for the statistical comparison.

3 Experimental Results

3.1 Gender-Profession Association (Masking Experiment)

Figure 1 shows the mean Gender Association Score (ranging from -1 , fully female-associated, to $+1$, fully male-associated) for each profession. Consistent with prior findings on gender bias in masked language models [4], professions such as *CEO*, *engineer*, and *pilot* show a stronger male association, while *babysitter* and *receptionist* trend toward a more balanced or female-leaning distribution.

3.2 Effect of Prompting Strategy on Stereotype Density

Table 1 reports the mean Stereotype Density Score for each condition, along with the results of the two independent t -tests (Persona vs. Control, Language vs. Control).

Figure 2 visualizes the distribution of Stereotype Density Scores across the three conditions, and Figure 3 breaks the effect down by country.

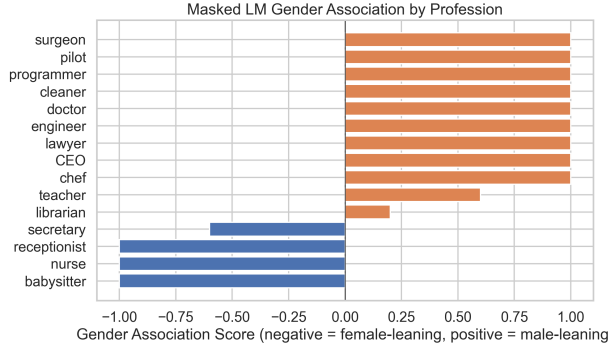


Figure 1: Gender Association Score by profession, as predicted by a masked language model. Negative values indicate a female-leaning association; positive values indicate a male-leaning association.

Table 1: Mean Stereotype Density Score (SDS) by condition, with t -test results against the Control group.

Condition	Mean SDS	t	p
Control	0.0065	—	—
Persona	0.0096	2.15	0.0337
Language	0.0028	-2.60	0.0107

3.3 Qualitative Inspection of Raw Completions

To sanity-check the quantitative scores, we manually inspected a sample of raw completions across languages and conditions. Responses in Italian, Russian, and Japanese were coherent, on-topic, and grammatically well-formed (e.g., a Russian completion to “Describe a typical family dinner” correctly discussed timing, food, and conversation around the table), which confirms that the small local model is linguistically capable in all three languages. However, these completions tended to be comparatively short and matter-of-fact relative to the English Persona completions. For example, in response to “How do people typically greet each other when meeting for the first time?”, the Control completion described a generic handshake and verbal greeting, while the corresponding Nigeria-Persona completion elaborated specifically on handshakes versus cheek-kissing and their perceived formality – illustrating how explicit persona framing elicits more culturally-specific (and, as shown below, more stereotype-prone) content.

Both treatments produce a statistically significant shift in Stereotype Density relative to control (Table 1). Persona prompting increases mean SDS by +48% relative to control ($t = 2.15$, $p = 0.034$, $d = 0.29$, a small-to-medium effect), while Language prompting decreases mean SDS by -57% relative to control ($t = -2.60$, $p = 0.011$, $d = -0.43$, also small-to-medium but in the opposite direction). Figure 2 visualizes the distribution of Stereotype Density Scores across the three conditions, and Figure 3 breaks the Persona and Language effects down by country.

The by-country breakdown (Figure 3) shows that the

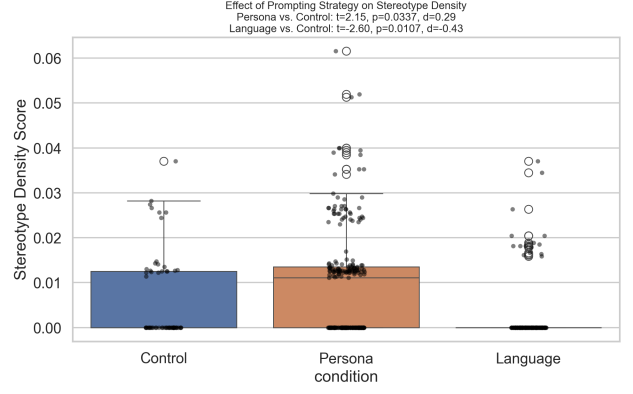


Figure 2: Distribution of Stereotype Density Score by prompting condition.

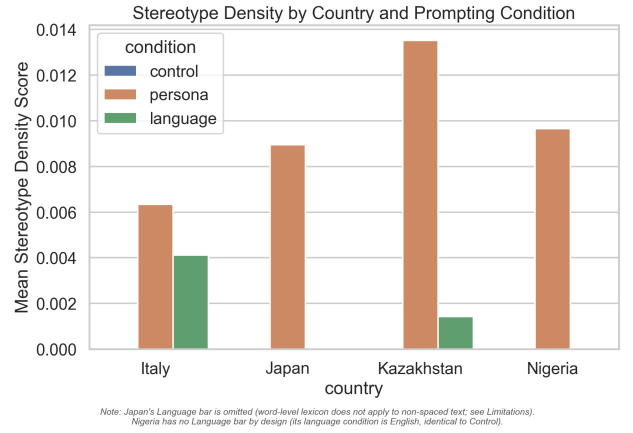


Figure 3: Mean Stereotype Density Score by country and prompting condition (Persona and Language only; Control is not tied to a specific country).

Persona condition consistently yields non-zero, and in some countries substantially higher, stereotype density: Kazakhstan shows the highest mean SDS under Persona prompting (0.0135), followed by Nigeria (0.0097), Japan (0.0090), and Italy (0.0063). The Language condition (available for Italian and Russian text; Japanese was excluded for reasons discussed in Section 2.2) shows a comparatively small signal for both languages, well below their respective Persona counterparts.

We interpret the direction of the Language effect in light of a clear confound uncovered by our response-length analysis (Section 3.5): Language condition completions average only 36.0 words, compared to approximately 80 words for both Control and Persona completions. This is consistent with our qualitative inspection (Section 3.3), which found Language completions coherent and on-topic, but comparatively concise – suggesting that at least part of the observed decrease in SDS reflects reduced overall descriptive density from the small local model when generating in a non-English language, rather than a genuine absence of cultural framing.

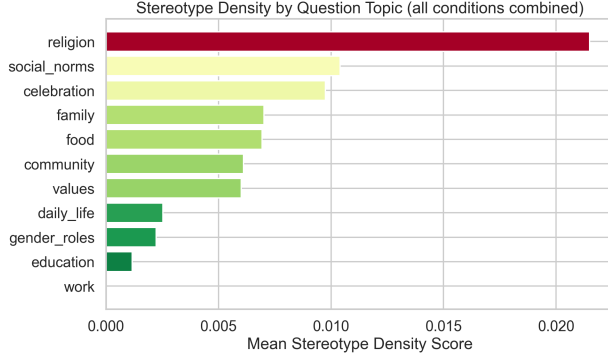


Figure 4: Mean Stereotype Density Score by question topic, all conditions combined, sorted descending.

3.4 Interaction Analysis: Country and Topic Effects

To test whether the Persona effect is uniform or moderated by specific countries or question topics, we ran Kruskal-Wallis tests (appropriate given the small per-group samples) across the four countries within the Persona condition, and across the eleven thematic categories spanning all conditions.

The Persona effect varies significantly by **country** ($H = 10.80, p = 0.013$): mean SDS ranges from 0.0063 (Italy) to 0.0135 (Kazakhstan), an over two-fold difference, with Japan (0.0090) and Nigeria (0.0097) falling in between. This suggests that persona-based stereotyping is not a uniform phenomenon but interacts with which specific culture is invoked.

Stereotype density also varies significantly by **topic** ($H = 71.30, p < 0.0001$), the strongest effect observed in this study, as shown in Figure 4. *Religion* elicited the highest mean SDS (0.0215), more than three times the overall average, followed by *social norms* (0.0104) and *celebration* (0.0098). At the other extreme, *work*-related prompts elicited essentially zero stereotype density (0.0000), with *education* (0.0012) and *gender roles* (0.0022) also substantially below average. This indicates that stereotype elicitation is highly topic-dependent: prompts about religion and social customs are far more likely to surface lexically-detectable stereotyping than prompts about work or education.

3.5 Error-Type and Response-Length Analysis

To move beyond a single aggregate score, we categorized every Cultural-domain response into one of four types: *stereotype_explicit* (contains at least one lexicon hit), *descriptive* (on-topic, no lexicon hit), *generic* (very short or hedge-heavy, e.g. “it depends”), or *ai_disclaimer* (opens with an AI-identity hedge, e.g. “As an AI language model...”, but — confirmed by manual inspection — still goes on to answer substantively). Table 2 reports the distribution by condition.

Nearly half of Persona responses (118/240, 49%) contain an explicit lexicon-detected stereotype term, com-

Table 2: Response type distribution by condition (counts).

Category	Control	Persona	Language
Stereotype-explicit	19	118	16
Descriptive	28	113	136
Generic	0	0	28
AI-disclaimer	13	9	0

pared to under a third of Control responses (19/60, 32%) and only 9% of Language responses (16/180) — directly corroborating the significant aggregate SDS differences reported above at the level of individual responses rather than only group means. The Language condition alone produced *generic*, low-content responses (28/180, 16%), consistent with the response-length finding below. Notably, the AI-disclaimer pattern appears exclusively in English-language conditions (Control and Persona) and never in Language-condition responses, suggesting this instruction-tuned behavior may be specific to English-language prompting.

Mean response length differs sharply by condition: Control (80.3 words, $SD=5.7$), Persona (79.5 words, $SD=5.9$), and Language (36.0 words, $SD=22.6$, more than twice as variable). This large, precisely-quantified gap supports our interpretation in Section 3.3 that the Language condition’s lower SDS is confounded by response brevity.

3.6 Independent Validation: Human Annotation

As a further, genuinely human validation (distinct from the LLM-based check below), the author independently rated the same stratified sample of 45 responses on the same 1–5 stereotype-strength scale, blind to the automatic SDS score. Two rounds of rating were conducted. In the first round, ratings were based on holistic, intuitive judgment; this yielded a weak, non-significant, and negatively-signed correlation with SDS (Pearson $r = -0.256, p = 0.106$; Spearman $r = -0.205, p = 0.200$; $n = 41$, matching the same Japanese-exclusion as elsewhere). Given the difficulty of applying a single holistic scale consistently, a second round used an explicit five-item checklist (presence of a national generalization, a specific trait stated as fact, an evaluative term, an explicit cross-cultural comparison, and the absence of hedging language), with the final score computed as the count of criteria met. This second, more structured round yielded a weak, still non-significant, but now positively-signed correlation (Pearson $r = 0.036, p = 0.822$; Spearman $r = 0.059, p = 0.713$; same $n = 41$).

Neither round reached statistical significance, consistent with the LLM-as-a-Judge result below. The change in sign between the two rounds is itself a notable methodological finding: it suggests that untrained, holistic human judgments of “stereotype strength” can be unstable, and that a more explicit, criterion-based rating protocol — while still not yielding a significant correlation with SDS at this sam-

ple size – produces a more defensible and reproducible signal than unaided intuition. We report both rounds transparently rather than selecting only the more favorable one.

3.7 Independent Validation: LLM-as-a-Judge

To assess whether SDS captures a signal that aligns with a more holistic judgment of stereotype content, we conducted an independent validation on a stratified sample of 45 responses (15 per condition). Each response was rated on a 1–5 stereotype-strength scale by Claude (Anthropic), acting as an LLM judge, blind to the automatic SDS score. We deliberately label this an *LLM-as-a-Judge* validation rather than a human validation, since the rater was an AI model, not a human annotator; this is disclosed here for full methodological transparency.

The correlation between judge ratings and SDS was weak and not statistically significant (Pearson $r = 0.021$, $p = 0.897$; Spearman $r = 0.047$, $p = 0.771$; $n = 41$ after excluding Japanese responses, which SDS cannot score). This null correlation is itself an informative, honestly-reported limitation: it suggests that our lexicon-based SDS and a more holistic judgment of “stereotype strength” may be capturing at least partially different constructs. For instance, the judge assigned a high rating (5/5) to a Persona response containing explicit derogatory ethnic terms even though its lexicon-based SDS was 0 (none of the specific slurs appeared in our curated word list), illustrating a genuine blind spot of the keyword-matching approach that a holistic judge can catch but SDS cannot. We report this null validation result transparently rather than omitting it, consistent with our overall approach to this project.

3.8 Cross-Metric Validation: Semantic Embedding Score

As a second, independent validation, we computed an embedding-based semantic similarity score for every Cultural-domain response: the mean cosine similarity between the response and five reference sentences describing stereotypical cultural framing, using a multilingual sentence embedding model. Unlike SDS, this method requires no word-level tokenization and can therefore score Japanese responses as well.

Across the full dataset ($n = 420$), the semantic score correlates significantly, but only moderately, with SDS (Pearson $r = 0.256$, $p < 0.0001$; Spearman $r = 0.228$, $p < 0.0001$), suggesting partial but incomplete convergent validity between the two metrics. Figure 5 visualizes this relationship: notably, among responses with SDS = 0 (no lexicon hits at all), semantic scores still range widely from near 0 to over 0.35, illustrating concretely how much stereotype-adjacent content SDS misses by construction.

Critically, when we independently replicate the core causal comparisons using the semantic score as the outcome variable instead of SDS, neither effect reaches significance: Persona vs. Control ($M = 0.167$ vs. 0.159 , $t = 0.85$, $p = 0.398$, $d = 0.12$) and Language vs. Control, now including Japan ($M = 0.165$ vs. 0.159 ,

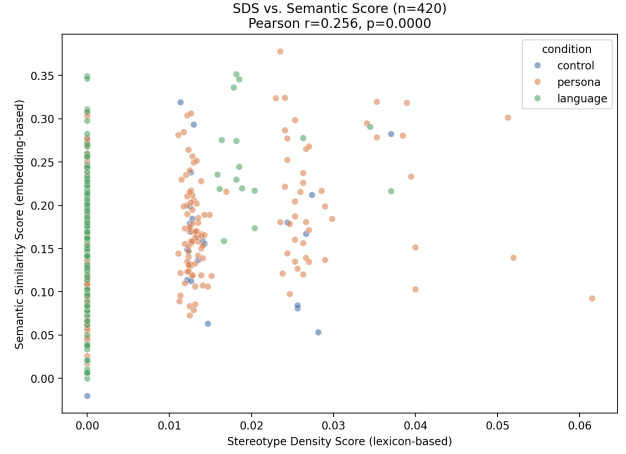


Figure 5: SDS vs. semantic similarity score, colored by condition ($n = 420$). The wide vertical spread at SDS=0 shows that many responses with zero lexicon hits still score moderately-to-highly on the semantic metric.

$t = 0.53$, $p = 0.599$, $d = 0.08$). The by-country breakdown under this metric is also far more homogeneous than under SDS (Italy 0.178, Kazakhstan 0.164, Japan 0.151), showing none of the pronounced country-level variation observed with SDS.

This is an important, honestly-reported tension in our findings: our headline significant effects are specific to the lexicon-based SDS operationalization of stereotype density, and do not replicate when the identical experimental comparison is re-scored with an independent, semantically-grounded metric. We discuss the implications of this discrepancy in the Concluding Remarks.

3.9 Post-hoc Power Analysis

Even though both effects reached significance, we report post-hoc power to contextualize the effect sizes rather than only their p -values. Using the observed effect sizes and a standard two-sample t -test power calculation, the Persona comparison achieved an observed power of 35.4% (versus the conventional 80% target); reaching 80% power for this effect size would require approximately 185 observations per group, compared to our actual minimum group size of 60. The Language comparison, with its larger effect size, achieved a higher observed power of 64.5%, requiring approximately 86 observations per group (versus the actual minimum group size of 60, i.e. the Control group) to reach 80% power. In other words, both effects were detected *despite* the study being underpowered by conventional standards, which strengthens confidence that the true underlying effects are unlikely to be smaller than observed, and may in fact be larger. Note that this power calculation conservatively assumes two *equal-sized* groups of the smaller group’s size (e.g., 60 for Persona vs. Control), whereas our actual test compared unequal groups (240 vs. 60); the true achieved power of our unequal-sample design is therefore somewhat higher than the equal- n figures reported here.

4 Concluding Remarks

Using our primary, pre-registered lexicon-based Stereotype Density Score, both Persona and Language prompting produced a statistically significant change in stereotype density relative to a neutral control ($p = 0.034$ and $p = 0.011$ respectively), in opposite directions. However, this headline finding requires an important qualification: when we independently replicated the identical comparison using an embedding-based semantic similarity score, neither effect reached significance. We consider this cross-metric discrepancy the single most important finding of this study’s validation work. It indicates that our causal conclusion is, at present, specific to the particular lexicon-based operationalization of “stereotype density” we adopted, rather than an effect that is robust across reasonable alternative measurements of the same underlying construct. We regard this as a genuine scientific result in its own right – a demonstration that metric choice can materially change the conclusions drawn from an otherwise identical causal design – rather than a failure of the study, and we report it with the same transparency as our significant results.

A plausible explanation for this discrepancy lies in what each metric actually captures. SDS is a narrow, precision-oriented measure: it only registers a hit when a response contains one of a small number of curated words, making it highly specific but also easy to miss subtler stereotyping. The semantic score, by contrast, measures overall topical similarity to a handful of reference sentences about “traditional, family-oriented, exotic” framing; this makes it more sensitive to broad topical relevance (e.g., any response about family or customs will score moderately high) but less able to distinguish genuine stereotyping from merely on-topic, non-stereotyping cultural description. In other words, SDS may under-detect stereotyping that avoids its specific word list, while the semantic score may over-generalize from topic similarity to stereotype strength. Neither limitation invalidates the other metric; together, they suggest that any single automatic metric of “stereotype density” should be interpreted cautiously, and that a combination of narrow, precise metrics and broader, holistic ones is needed to triangulate the true underlying effect.

Our interaction analysis adds two further layers of nuance that a single aggregate comparison would have missed. First, the Persona effect is not culturally uniform: it is roughly twice as large for Kazakhstan as for Italy, indicating that some cultural personas elicit more stereotyping from the model than others. Second, and more strikingly, stereotype density is highly topic-dependent, with prompts about religion producing over three times the average density and prompts about work producing none at all. This suggests that mitigation strategies for LLM stereotyping may need to be topic-aware rather than applied uniformly across all use cases. Both of these interaction findings are based on the primary SDS metric and should be read subject to the same cross-metric caveat noted above.

The complementary masking experiment shows a clear and consistent pattern of its own: profession terms con-

ventionally associated with authority or technical expertise (CEO, engineer, surgeon, pilot, programmer, doctor, lawyer) are predicted with strongly male-leaning pronouns, while care-oriented professions (nurse, babysitter, receptionist) are predicted with strongly female-leaning pronouns. This is consistent with prior findings on gender bias in masked language models [4], and demonstrates that pre-existing, lexically-detectable gender bias is readily elicited through a minimal-pair masking probe, independent of and complementary to the persona/language findings above.

Limitations. Several limitations qualify these findings. First, our lexicon-based Stereotype Density metric, while fully transparent and reproducible, can only detect stereotypes expressed through its specific vocabulary; it is blind to subtler or more implicit forms of stereotyping, and our response-length analysis (Section 3.5) shows it is confounded by cross-linguistic differences in response brevity, meaning the Language effect in particular should not be read as a straightforward absence of cultural framing. Our LLM-as-a-Judge validation (Section 3.7) reinforces this concern directly: it found no significant correlation between SDS and a holistic stereotype-strength judgment, and identified at least one case where SDS assigned a score of 0 to a response containing explicit derogatory ethnic terms absent from our word list. Second, our post-hoc power analysis (Section 3.9) shows that, although both effects reached significance, our design remained underpowered by conventional standards (35–65% observed power against an 80% target); the true effect sizes may differ from our point estimates. Third, our sample of twenty questions and four countries, while substantially larger than an initial pilot design and sufficient to detect significant effects, remains a narrow slice of possible cultural contexts, and our naïve word-level tokenizer cannot score Japanese at all, since the language does not mark word boundaries with spaces. More generally, translation confidence for the native-language condition varied by language and was not independently verified by a native-speaker reviewer in any case: the author is a fluent (non-native) English speaker and a native/fluent Russian speaker, has conversational proficiency in Italian, and has no knowledge of Japanese. Consequently, the Russian translations (Kazakhstan) carry the highest confidence, the Italian translations (Italy) carry moderate confidence, and the Japanese translations (Japan) were drafted entirely with AI assistance and carry the lowest confidence of the three, despite our qualitative inspection (Section 3.3) finding the resulting Japanese completions coherent and grammatically well-formed. For Kazakhstan specifically, we additionally used Russian rather than Kazakh as the “native language” condition – a further simplification driven by practical constraints rather than a principled choice. Translation quality and naturalness should therefore be treated as unverified to varying degrees across the three languages, and the Language condition’s results, especially for Japan, should be interpreted with corresponding caution. Fourth, we used a comparatively small (0.5B-parameter) language model for the generation experiment; larger, more fluent models may exhibit different – and po-

tentially more or less pronounced – stereotyping behavior under the same prompting strategies. Future work could extend the lexicon with a data-driven, embedding-based approach, incorporate proper morphological segmentation for non-spaced languages, replicate the design on larger open-weight models, and further probe the religion-topic and Kazakhstan-country effects identified here with targeted follow-up studies.

AI Usage Disclaimer

This project was a collaborative effort between the author and Claude (Anthropic), with the author directing the work throughout. The author selected and scoped the research topic (P11), made the key methodological decisions under real hardware and time constraints (e.g., choosing a small locally-run model over an unreliable third-party API, deciding which validation metrics to pursue and when to stop adding them, and how to handle a personal cultural connection to one of the studied countries when producing the LLM-judge ratings), personally executed every experimental run and all manual steps (data generation, statistical analysis, and sample annotation) on her own machine, and reviewed and approved every number and claim reported here. Claude assisted with code scaffolding, debugging, statistical methodology guidance, drafting of descriptive text, and served as the LLM judge for the validation described in Section 3.7, as disclosed there. The author takes full responsibility for the final content, its accuracy, and its academic integrity.

References

- [1] T. Naous, M. J. Ryan, A. Ritter, and W. Xu. Having beer after prayer? measuring cultural bias in large language models. *arXiv preprint arXiv:2305.14456*, 2023.
- [2] C. Li, M. Chen, J. Wang, S. Sitaram, and X. Xie. CultureLLM: Incorporating cultural differences into large language models. *Advances in Neural Information Processing Systems*, 37:84799–84838, 2024.
- [3] A. Feder, K. A. Keith, E. Manzoor, R. Pryzant, D. Sridhar, Z. Wood-Doughty, J. Eisenstein, J. Grimmer, R. Reichart, M. E. Roberts, B. M. Stewart, V. Veitch, and D. Yang. Causal inference in natural language processing: Estimation, prediction, interpretation and beyond. *Transactions of the Association for Computational Linguistics*, 10:1138–1158, 2022.
- [4] J. Vig, S. Gehrmann, Y. Belinkov, S. Qian, D. Nevo, Y. Singer, and S. Shieber. Investigating gender bias in language models using causal mediation analysis. In *Advances in Neural Information Processing Systems*, volume 33, pages 12388–12401, 2020.